第二章：量子动力学

§2.1 薛定谔方程

$$i\hbar\frac{d}{dt}|\psi(t)\rangle = \hat{H}(t)|\psi(t)\rangle$$

$\hat{H}(t)$：哈密顿算符

- 方程的基本性质

(a) 关于 t 的一阶微分方程.

设初态：$|\psi(t_0)\rangle$ 则

$|\psi(t)\rangle$ 由薛定谔方程唯一确定. $(t>t_0)$

(b) 方程是线性齐次的，迭加原理成立.

设 $|\psi_1(t)\rangle$ 和 $|\psi_2(t)\rangle$ 为方程的解：

$$\begin{cases} i\hbar\frac{d}{dt}|\psi_1(t)\rangle = \hat{H}(t)|\psi_1(t)\rangle \\ i\hbar\frac{d}{dt}|\psi_2(t)\rangle = \hat{H}(t)|\psi_2(t)\rangle \end{cases}$$ 则

$$|\psi(t)\rangle = \lambda_1 |\psi_1(t)\rangle + \lambda_2 |\psi_2(t)\rangle$$

也是方程的解！ （λ_1, λ_2为复常数）

证明：

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \lambda_1 \left(i\hbar \frac{\partial}{\partial t} |\psi_1(t)\rangle \right)$$

$$+ \lambda_2 \left(i\hbar \frac{\partial}{\partial t} |\psi_2(t)\rangle \right)$$

$$= \lambda_1 \left(\hat{H}(t) |\psi_1(t)\rangle \right)$$

$$+ \lambda_2 \left(\hat{H}(t) |\psi_2(t)\rangle \right)$$

$$= \hat{H}(t) \left[\lambda_1 |\psi_1(t)\rangle + \lambda_2 |\psi_2(t)\rangle \right]$$

$$= \hat{H}(t) |\psi(t)\rangle.$$

证毕

- $|\psi(t_0)\rangle = \lambda_1 |\psi_1(t_0)\rangle + \lambda_2 |\psi_2(t_0)\rangle$

(二) 粒子数守恒. (非相对论量子力学)

- 方程的解 $|\psi(t)\rangle$ 满足 方程：

$$\frac{d}{dt}\langle\psi(t)|\psi(t)\rangle = 0 .$$

证明：

$$\begin{cases} \frac{d}{dt}|\psi(t)\rangle = \frac{1}{i\hbar}\hat{H}(t)|\psi(t)\rangle \\ \frac{d}{dt}\langle\psi(t)| = -\frac{1}{i\hbar}\langle\psi(t)|\hat{H}(t) \end{cases}$$

$$\Rightarrow \frac{d}{dt}\langle\psi(t)|\psi(t)\rangle = \left(\frac{d}{dt}\langle\psi(t)|\right)|\psi(t)\rangle + \langle\psi(t)|\left(\frac{d}{dt}|\psi(t)\rangle\right)$$

$$= -\frac{1}{i\hbar}\langle\psi(t)|\hat{H}(t)|\psi(t)\rangle + \langle\psi(t)|\frac{1}{i\hbar}\hat{H}(t)|\psi(t)\rangle$$

$$= 0 .$$

证毕

物理含意

- $\langle \psi(t) | \psi(t) \rangle = \int dx \ \langle \psi(t) | x \rangle \langle x | \psi(t) \rangle$

$= \int dx \ \psi^*(t,x) \ \psi(t,x)$

$= \int dx \ |\psi(t,x)|^2$ （t时刻在整个实空间找到粒子的概率）

- $\langle \psi(t_0) | \psi(t_0) \rangle$

$= \int dx \, |\psi(t_0, x)|^2$ （t_0时刻在整个实空间找到粒子的概率）

所 $\frac{d}{dt} \langle \psi(t) | \psi(t) \rangle = 0$

$\Rightarrow \langle \psi(t) | \psi(t) \rangle = \langle \psi(t_0) | \psi(t_0) \rangle$

$\Rightarrow \int dx \, |\psi(t,x)|^2 = \int dx \, |\psi(t_0, x)|^2$.

即 $\int dx \, |\psi(t_0, x)|^2 = 1$. 以对任意时刻 t，有

$$\int dx \, |\psi(t, x)|^2 = 1$$

§2.2 时间演化算符：$\hat{U}(t,t_0)$

time evolution operator

(a) 定义式

$$|\psi(t)\rangle \equiv \hat{U}(t,t_0)\ |\psi(t_0)\rangle$$

(b) 若干性质

- 令 $t=t_0$ ，

$$|\psi(t_0)\rangle = \hat{U}(t_0,t_0)\,|\psi(t_0)\rangle$$

$$\Rightarrow \quad \underline{\hat{U}(t_0,t_0)=\mathbb{1}}$$

$$\Rightarrow \quad \underline{\hat{U}(t,t)=\mathbb{1}}$$

$\because |\psi(t)\rangle = \hat{U}(t,t_0)\,|\psi(t_0)\rangle$ 所对应左矢：

$$\langle\psi(t)| = \langle\psi(t_0)|\,\hat{U}^{\dagger}(t,t_0) \qquad (\hat{U}^{\dagger}(t,t) = \mathbb{1})$$

一方面：$\langle\psi(t)|\psi(t)\rangle =$

$$\langle\psi(t_0)|\,\hat{U}^{\dagger}(t,t_0)\,\hat{U}(t,t_0)\,|\psi(t_0)\rangle \,,$$

另一方面：$\langle\psi(t)|\psi(t)\rangle = \langle\psi(t_0)|\psi(t_0)\rangle$ (概率守恒)

$$\Rightarrow \quad \hat{U}^{\dagger}(t,t_0)\,\hat{U}(t,t_0) = \mathbb{1} \qquad \cdots(a)$$

$\because |\psi(t)\rangle = \hat{U}(t,t_0)\,|\psi(t_0)\rangle$，左乘 $\hat{U}^{\dagger}(t,t_0)$

$$\Rightarrow \quad \hat{U}^{\dagger}(t,t_0)\,|\psi(t)\rangle = \hat{U}^{\dagger}(t,t_0)\,\hat{U}(t,t_0)\,|\psi(t_0)\rangle$$

$$= |\psi(t_0)\rangle$$

对称无关：

$$\langle\psi(t_0)| = \langle\psi(t)|\hat{U}(t,t_0)$$

一方面：$\langle\psi(t_0)|\psi(t_0)\rangle =$

$$\langle\psi(t)|\hat{U}(t,t_0)\,\hat{U}^{\dagger}(t,t_0)|\psi(t)\rangle$$

另一方面：$\langle\psi(t_0)|\psi(t_0)\rangle = \langle\psi(t)|\psi(t)\rangle$

$$\Rightarrow \quad \hat{U}(t,t_0)\,\hat{U}^{+}(t,t_0) = \mathbb{1} \qquad \cdots (b)$$

由 (a), (b)

$$\Rightarrow \quad \boxed{\hat{U}^{+}(t,t_0) = \hat{U}^{-1}(t,t_0)} \qquad (\text{幺正性})$$

时间演化算符 $\hat{U}(t,t_0)$ 是幺正算符！

✱ 设 $t_2 > t_1 > t_0$.

一方面： $|\psi(t_2)\rangle = \hat{U}(t_2, t_0)\,|\psi(t_0)\rangle$

另一方面： $|\psi(t_2)\rangle = \hat{U}(t_2, t_1)\,|\psi(t_1)\rangle$

$$= \hat{U}(t_2, t_1)\,\hat{U}(t_1, t_0)\,|\psi(t_0)\rangle$$

两者相同 $\Rightarrow$

$$\hat{U}(t_2, t_0)\,|\psi(t_0)\rangle = \hat{U}(t_2, t_1)\,\hat{U}(t_1, t_0)\,|\psi(t_0)\rangle$$

$|\psi(t_0)\rangle$ 任意， $\Rightarrow$

$$\boxed{\hat{U}(t_2, t_0) = \hat{U}(t_2, t_1)\,\hat{U}(t_1, t_0)}$$

（传递性）

时间演化算符的性质（续）

$$\hat{U}(t_0, t_1) = \hat{U}^{-1}(t_1, t_0)$$

证明：由时间演化算符的传递性，

有 $\hat{U}(t_2, t_1)\hat{U}(t_1, t_0) = \hat{U}(t_2, t_0)$ 。

上式中令 $t_2 = t_0$ 得，

$$\hat{U}(t_0, t_1)\hat{U}(t_1, t_0) = \mathbb{1} \quad \cdots(1)$$

以此中交换 t_0 与 t_1，得

$$\hat{U}(t_1, t_0)\hat{U}(t_0, t_1) = \mathbb{1} \quad \cdots(2)$$

$$\Rightarrow \quad \hat{U}(t_0, t_1) = \hat{U}^{-1}(t_1, t_0) \ .$$

$$\text{or} \quad \hat{U}(t_0, t_1) = \hat{U}^{\dagger}(t_1, t_0)$$

(C) $\hat{U}(t,t_0)$ 的具体表达式

- 随时间演化

$$\begin{cases} i\hbar\frac{d}{dt}|\psi(t)\rangle = \hat{H}(t)|\psi(t)\rangle \\ |\psi(t)\rangle = \hat{U}(t,t_0)|\psi(t_0)\rangle \end{cases}.$$

$$\Rightarrow \quad i\hbar\frac{d}{dt}\left[\hat{U}(t,t_0)|\psi(t_0)\rangle\right] = \hat{H}(t)\left[\hat{U}(t,t_0)|\psi(t_0)\rangle\right].$$

$$\Rightarrow \quad \left[i\hbar\frac{d}{dt}\hat{U}(t,t_0)\right]|\psi(t_0)\rangle = \left[\hat{H}(t)\hat{U}(t,t_0)\right]|\psi(t_0)\rangle$$

$|\psi(t_0)\rangle$ 任意 $\Rightarrow$

$$\boxed{i\hbar\frac{d}{dt}\hat{U}(t,t_0) = \hat{H}(t)\hat{U}(t,t_0)}$$

初始条件：$\hat{U}(t,t)\big|_{t=t_0} = \mathbb{1}$

- $\hat{U}(t,t_0)$ 的积分表达式

- $i\hbar \frac{d}{dt}\hat{U}(t,t_0) = \hat{H}(t)\hat{U}(t,t_0)$

- 上方程两边对时间积分：（上限为 t，下限为 t_0）

$$\int_{t_0}^{t} dt' \frac{d}{dt'}\hat{U}(t',t_0) = \frac{1}{i\hbar}\int_{t_0}^{t} dt' \hat{H}(t')\hat{U}(t',t_0) \quad ,$$

$$\hat{U}(t,t_0) - \hat{U}(t_0,t_0) = \frac{1}{i\hbar}\int_{t_0}^{t} dt' \hat{H}(t')\hat{U}(t',t_0) \quad ,$$

$\hat{U}(t_0,t_0) = \mathbb{1}$

$$\Rightarrow \quad \boxed{\hat{U}(t,t_0) = \mathbb{1} + \frac{1}{i\hbar}\int_{t_0}^{t} dt' \hat{H}(t')\hat{U}(t',t_0)}$$

（积分方程）

满足初始条件

$\hat{U}(t_0,t_0) = \mathbb{1}$

$$\hat{U}(t,t_0) = \mathbb{1} + \frac{1}{i\hbar}\int_{t_0}^{t} dt'\ \hat{H}(t')\hat{U}(t',t_0)$$

利用迭代法：

$$\begin{cases} \hat{U}_n(t,t_0) = \mathbb{1} + \dfrac{1}{i\hbar}\displaystyle\int_{t_0}^{t} \hat{H}(t')\hat{U}_{n-1}(t',t_0) & (n\geq 1) \\ \hat{U}_0(t,t_0) = \mathbb{1} \end{cases}$$

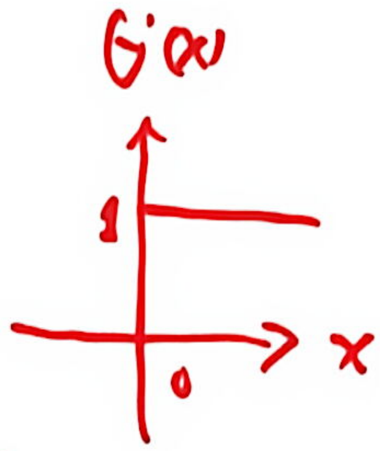
$$\hat{U}(t,t_0) = \mathbb{1} + \sum_{n\geq 1}^{\infty}\left(\frac{1}{i\hbar}\right)^n \int_{t_0}^{t} dt_1 \int_{t_0}^{t_1} dt_2 \cdots \int_{t_0}^{t_{n-1}} dt_n\, \hat{H}(t_1)\hat{H}(t_2) \cdots \hat{H}(t_n)$$

$$\left(t_0 \leq t_n \leq t_{n-1} \cdots \leq t_1 \leq t \right)$$

- 编时算符 $\hat{T}$

$$\hat{T}\left[\hat{H}(t_1)\hat{H}(t_2)\right] \equiv \begin{cases} \hat{H}(t_1)\hat{H}(t_2), & t_1 > t_2 \\ \hat{H}(t_2)\hat{H}(t_1), & t_2 > t_1 \end{cases}$$

- 阶跃函数 $\theta(x) \equiv \begin{cases} 1, & x > 0 \\ 0, & x < 0 \end{cases}$



$$\left(\theta(x) + \theta(-x) = 1\right)$$

$$\Rightarrow \quad \hat{T}\left[\hat{H}(t_1)\hat{H}(t_2)\right] = \theta(t_1 - t_2)\,\hat{H}(t_1)\hat{H}(t_2) + \theta(t_2 - t_1)\,\hat{H}(t_2)\hat{H}(t_1)$$

$\hat{T}\left[\hat{H}(t_1)\hat{H}(t_2)\hat{H}(t_3)\right] =$ 6 项之和.

(相互)

$$\hat{T}\left[\hat{H}(t_1)\hat{H}(t_2)\right] = \hat{T}\left[\hat{H}(t_2)\hat{H}(t_1)\right]$$

• 第三项：

$$I_3 = \frac{1}{(i\hbar)^2}\int_{t_0}^{t} dt_1 \int_{t_0}^{t_1} dt_2 \; \hat{H}(t_1)\hat{H}(t_2) \;,$$

$$(\; t_1 \longleftrightarrow t_2 \;)$$

$$I_3 = \frac{1}{(i\hbar)^2}\int_{t_0}^{t} dt_2 \int_{t_0}^{t_2} dt_1 \; \hat{H}(t_2)\hat{H}(t_1)$$

$$\Rightarrow \quad I_3 = \frac{1}{2!(i\hbar)^2}\left[\int_{t_0}^{t} dt_1 \int_{t_0}^{t_1} dt_2 \; \hat{H}(t_1)\hat{H}(t_2) + \int_{t_0}^{t} dt_2 \int_{t_0}^{t_2} dt_1 \; \hat{H}(t_2)\hat{H}(t_1)\right].$$

注，$\int_{t_0}^{t_1} dt_2 \; \hat{H}(t_1)\hat{H}(t_2) = \int_{t_0}^{t} dt_2 \; \theta(t_1-t_2)\hat{H}(t_1)\hat{H}(t_2)$

$\because$ $\int_{t_0}^{t_2} dt_1 \; \hat{H}(t_2)\hat{H}(t_1) = \int_{t_0}^{t} dt_1 \; \theta(t_2-t_1)\hat{H}(t_2)\hat{H}(t_1)$

$$\Rightarrow \quad I_3 = \frac{1}{2!(i\hbar)^2}\left[\int_{t_0}^{t} dt_1 \int_{t_0}^{t} dt_2\, \theta(t_1-t_2)\,\hat{H}(t_1)\hat{H}(t_2)\right.$$

$$\left. + \int_{t_0}^{t} dt_2 \int_{t_0}^{t} dt_1\, \theta(t_2-t_1)\,\hat{H}(t_2)\hat{H}(t_1)\right]$$

$$= \frac{1}{2!(i\hbar)^2}\int_{t_0}^{t} dt_1 \int_{t_0}^{t} dt_2\, \hat{T}\left[\hat{H}(t_1)\hat{H}(t_2)\right]$$

$$\Rightarrow \quad I_{n+1} = \frac{1}{n!(i\hbar)^n}\int_{t_0}^{t} dt_1 \int_{t_0}^{t} dt_2 \cdots \int_{t_0}^{t} dt_n\, \hat{T}\left[\hat{H}(t_1)\cdots\hat{H}(t_n)\right].$$

$$\hat{U}(t,t_0) = \mathbb{1} + \sum_{n=1}^{\infty} \frac{1}{n!(i\hbar)^n}\int_{t_0}^{t} dt_1 \int_{t_0}^{t} dt_2 \cdots \int_{t_0}^{t} dt_n$$

$$\times\ \hat{T}\left[\hat{H}(t_1)\hat{H}(t_2)\cdots\hat{H}(t_n)\right]$$

$$= \hat{T}\exp\left[\frac{1}{i\hbar}\int_{t_0}^{t} dt'\,\hat{H}(t')\right]$$

- 不同时间的 $\hat{H}$ 相互对易情况 (利用 $\theta(x)+\theta(-x)=1$)

$$\hat{U}(t,t_0)=\exp\left[\frac{1}{i\hbar}\int_{t_0}^{t}dt'\hat{H}(t')\right].$$

- $\hat{H}$ 与时间无关

$$\hat{U}(t,t_0)=\exp\left(\frac{\hat{H}(t-t_0)}{i\hbar}\right)$$

$$=\exp\left[-\frac{i\hat{H}(t-t_0)}{\hbar}\right]$$

(3) 例: $|\psi(t_0)\rangle=|\psi(t=0)\rangle=|\alpha\rangle$.

则 $|\psi(t)\rangle=\hat{U}(t,0)|\alpha\rangle$

$$=\exp\left[-\frac{i}{\hbar}\hat{H}t\right]|\alpha\rangle$$

$$=e^{-i\omega t/2}|\alpha e^{-i\omega t}\rangle\equiv|\alpha,t\rangle.$$

(b) 当 $\hat{H}$ 不显含时间时，有

$$\hat{U}(t,t_0) = \exp\left[-\frac{i}{\hbar}\hat{H}(t-t_0)\right].$$

设 $\hat{H}$ 本征矢与本征为

$$\hat{H}|E_n\rangle = E_n|E_n\rangle .$$

$$\begin{cases} \langle E_{n'}|E_n\rangle = \delta_{n'n} \\ \sum_n |E_n\rangle\langle E_n| = \mathbb{1}. \end{cases}$$

可得 $\hat{U}(t,t_0)$ 的矩表示：

$$\hat{U}(t,t_0) = \mathbb{1}\,\hat{U}(t,t_0)\,\mathbb{1}$$

$$= \sum_n |E_n\rangle\langle E_n|\,\hat{U}(t,t_0)\sum_{n'}|E_{n'}\rangle\langle E_{n'}|$$

$$= \sum_{n'}\sum_n |E_n\rangle\langle E_n|\exp\left[-\frac{i}{\hbar}\hat{H}(t-t_0)\right]|E_{n'}\rangle\langle E_{n'}|$$

$$= \sum_{n'} \sum_{n} |E_n\rangle\langle E_n|E_{n'}\rangle\langle E_{n'}| \exp\left[\frac{-i}{\hbar} E_{n'}(t-t_0)\right]$$

$$= \underline{\sum_n \exp\left[-\frac{iE_n(t-t_0)}{\hbar}\right] |E_n\rangle\langle E_n|} \quad .$$

. $\underline{|\psi(t)\rangle} = \hat{U}(t,t_0)\,|\psi(t_0)\rangle$

$$= \sum_n \exp\left[-\frac{iE_n(t-t_0)}{\hbar}\right] |E_n\rangle \underbrace{\langle E_n|\psi(t_0)\rangle}_{C_n}$$

$$= \underline{\sum_n C_n \exp\left[-\frac{iE_n(t-t_0)}{\hbar}\right] |E_n\rangle} \quad .$$

. $\underline{\psi(x,t)} = \langle x|\psi(t)\rangle = \sum_n C_n \exp\left[-\frac{iE_n(t-t_0)}{\hbar}\right] \underbrace{\langle x|E_n\rangle}_{\psi_n(x)}$

$$= \underline{\sum_n C_n \exp\left[-\frac{iE_n(t-t_0)}{\hbar}\right] \psi_n(x)}$$

(c) 증명 : $[\hat{H}(t_1), \hat{H}(t_2)] \neq 0$

$$\hat{H}(t) = \frac{\hat{p}^2}{2m} - A\sin\omega t\ \hat{x}$$

의 $[\hat{H}(t_1), \hat{H}(t_2)]$

$$= \left[\frac{\hat{p}^2}{2m} - A\sin\omega t_1 \hat{x},\ \frac{\hat{p}^2}{2m} - A\sin\omega t_2 \hat{x}\right]$$

$$= \left[\frac{\hat{p}^2}{2m}, -A\sin\omega t_2 \hat{x}\right] + \left[-A\sin\omega t_1 \hat{x}, \frac{\hat{p}^2}{2m}\right]$$

$$= -A\sin\omega t_2 \left[\frac{\hat{p}^2}{2m}, \hat{x}\right] - A\sin\omega t_1 \left[\hat{x}, \frac{\hat{p}^2}{2m}\right]$$

$$= \left(-A\sin\omega t_2 + A\sin\omega t_1\right)\left[\frac{\hat{p}^2}{2m}, \hat{x}\right]$$

$$= \frac{-A}{2m}\left[\sin\omega t_2 - \sin\omega t_1\right]\left[\hat{p}^2, \hat{x}\right]$$

$$= \frac{2\hbar A}{2m}\left[\sin\omega t_2 - \sin\omega t_1\right] \neq 0.$$

§2.3 量子力学中的三个绘景

薛定谔绘景 Schrödinger

海森伯绘景 Heisenberg

相互作用绘景 Interaction (Dirac)

绘景：量子系统随时间变化的表达方式

(1) 薛定谔绘景

(a) 态矢量：$|\psi(t)\rangle$

$$\begin{cases} i\hbar\dfrac{d}{dt}|\psi(t)\rangle=\hat{H}(t)|\psi(t)\rangle & \text{(薛定谔方程)} \\ \text{初始态：}|\psi(t_0)\rangle \end{cases}$$

- $|\psi(t)\rangle=\hat{U}(t,t_0)|\psi(t_0)\rangle$

$$\begin{cases} i\hbar\dfrac{d}{dt}\hat{U}(t,t_0)=\hat{H}\hat{U}(t,t_0) \\ \text{初始条件 } \hat{U}(t_0,t_0)=\mathbb{1} \end{cases}$$

- $\hat{H}(t_i)$ 间互对易或 $\hat{H}(t)=\hat{H}$ 时 $\hat{U}(t,t_0)$ 简单

(b) 算符 : $\hat{A}(t)$

常见力学量算符 : $\hat{x}$、$\hat{p}$、$\hat{L}$、$\hat{S}$ 等

都不显含时间

$\Rightarrow$ $\frac{d}{dt}\hat{A}(t)=0$ $\left(\hat{A}(t)=\hat{A}\right)$

(C) 算符平均值及平均值随时间变化

· $\overline{\hat{A}_t} = \langle \psi(t) | \hat{A} | \psi(t) \rangle$

力学量 ; └→任意态矢量

· $i\hbar \frac{d}{dt} \overline{\hat{A}_t} = \left(i\hbar \frac{d}{dt} \langle \psi(t) | \right) \hat{A} | \psi(t) \rangle$

$+ \langle \psi(t) | \hat{A} \left(i\hbar \frac{d}{dt} | \psi(t) \rangle \right)$

$= \langle \psi(t) | [\hat{A}, \hat{H}(t)] | \psi(t) \rangle$

(利用 : $-i\hbar \frac{d}{dt} \langle \psi(t) | = \langle \psi(t) | \hat{H}(t)$)

若 $[\hat{A}, \hat{H}(t)] = 0$

则 $\frac{d}{dt}\overline{\hat{A}_t} = 0$.

称力学量 A 为守恒量. 可证力学量 A 在任意态下的概率分布不随时间改变!

(d) A 一表象.

$$\hat{A}|a_i\rangle = a_i|a_i\rangle \qquad i=1,2,\cdots$$

$\{|a_i\rangle\}$ 基矢 (不显含时间)

$$\begin{cases} \langle a_i|a_j\rangle = \delta_{ij} \\ \sum_i |a_i\rangle\langle a_i| = \mathbb{1} \end{cases}$$

$$|\psi(t)\rangle = \sum_i |a_i\rangle\langle a_i|\psi(t)\rangle$$

* 基矢不随时间变化, 体态矢则随时间变化!

例：　求平均值 $\overline{\hat{x}_t} \equiv \langle \alpha, t | \hat{x} | \alpha, t \rangle$.

设 $t=0$ 时为态为 $|\alpha\rangle$.　$\left(|\psi(t)\rangle = |\alpha, t\rangle\right)$

先利用时间演化算符求 $|\alpha, t\rangle$.

- $|\alpha, t\rangle = \exp\left[-\frac{i}{\hbar}\hat{H}t\right]|\alpha\rangle = e^{-i\omega t/2}|\alpha e^{-i\omega t}\rangle$.

- $\overline{\hat{x}_t} = \langle \alpha e^{-i\omega t}| e^{i\omega t/2}\hat{x}\, e^{-i\omega t/2}|\alpha e^{-i\omega t}\rangle$

$$= \langle \alpha e^{-i\omega t}|\hat{x}|\alpha e^{-i\omega t}\rangle$$

$$= \sqrt{\frac{\hbar}{2m\omega}}\langle \alpha e^{-i\omega t}|(\hat{a}^\dagger + \hat{a})|\alpha e^{-i\omega t}\rangle$$

$$= \sqrt{\frac{\hbar}{2m\omega}}\left(\alpha^* e^{i\omega t} + \alpha e^{-i\omega t}\right)$$

$$= |\alpha|\sqrt{\frac{2\hbar}{m\omega}}\cos(\omega t - \phi).$$

$\left(\alpha = |\alpha| e^{i\phi}\right)$

(2) 海森伯绘景

(a) 态矢量的定义

→ 薛定谔绘景态矢

- $|\psi_H(t)\rangle \equiv \hat{U}^\dagger(t,t_0)|\psi(t)\rangle$

令 $t=t_0$

$|\psi_H(t_0)\rangle = \hat{U}^\dagger(t_0,t_0)|\psi(t_0)\rangle$

$= |\psi(t_0)\rangle$

* 两个绘景态矢量在 $t=t_0$ 时刻重合！

- $|\psi_H(t)\rangle$ 的时间演化

$$i\hbar\frac{d}{dt}|\psi_H(t)\rangle = i\hbar\frac{d}{dt}\left[\hat{U}^\dagger(t,t_0)|\psi(t)\rangle\right]$$

$$= \left(i\hbar \frac{d}{dt} \hat{U}^{\dagger}(t,t_0)\right) |\psi(t)\rangle + \hat{U}^{\dagger}(t,t_0) \left[i\hbar \frac{d}{dt} |\psi(t)\rangle\right]$$

$$= -\hat{U}^{\dagger}(t,t_0) \hat{H}(t) |\psi(t)\rangle + \hat{U}^{\dagger}(t,t_0) \hat{H}(t) |\psi(t)\rangle$$

$$= 0$$

即

$$\frac{d}{dt} |\psi_H(t)\rangle = 0$$

* 海森伯绘景中，态矢量不随时间变化！

(b) 算符的定义

薛定谔绘景算符

$$\hat{A}_H(t) \equiv \hat{U}^{\dagger}(t,t_0) \hat{A} \hat{U}(t,t_0)$$

- $\hat{A}_H(t) = \hat{U}^\dagger(t,t_0)\,\hat{A}\,\hat{U}(t,t_0)$

  令 $t = t_0 \Rightarrow \hat{A}_H(t_0) = \hat{A}$

  两者重合！

- $\hat{A}_H(t)$ 的时间演化

$$i\hbar\frac{d}{dt}\hat{A}_H(t) = i\hbar\frac{d}{dt}\left[\hat{U}^\dagger(t,t_0)\,\hat{A}\,\hat{U}(t,t_0)\right]$$

$$= \left(i\hbar\frac{d}{dt}\hat{U}^\dagger(t,t_0)\right)\hat{A}\,\hat{U}(t,t_0)$$

$$+ \hat{U}^\dagger(t,t_0)\,\hat{A}\left(i\hbar\frac{d}{dt}\hat{U}(t,t_0)\right)$$

$$= -\hat{U}^\dagger(t,t_0)\,\hat{H}(t)\,\hat{A}\,\hat{U}(t,t_0)$$

$$+ \hat{U}^\dagger(t,t_0)\,\hat{A}\,\hat{H}(t)\,\hat{U}(t,t_0)$$

$$= -\hat{U}^{\dagger}(t,t_0)\,\hat{H}(t)\,\underbrace{\hat{U}(t,t_0)\,\hat{U}^{\dagger}(t,t_0)}_{=1}\,\hat{A}\,\hat{U}(t,t_0)$$

$$+\hat{U}^{\dagger}(t,t_0)\,\hat{A}\,\underbrace{\hat{U}(t,t_0)\,\hat{U}^{\dagger}(t,t_0)}_{=1}\,\hat{H}\,\hat{U}(t,t_0)$$

$$= -\hat{H}_H(t)\,\hat{A}_H(t)$$

$$+\hat{A}_H(t)\,\hat{H}_H(t) = \left[\hat{A}_H(t),\hat{H}_H(t)\right].$$

即

$$\boxed{i\hbar\frac{d\hat{A}_H(t)}{dt} = \left[\hat{A}_H(t),\hat{H}_H(t)\right]}$$

海森伯方程.

初始条件：$\hat{A}_H(t_0)=\hat{A}$

(C) 平均值与守恒量

- $\overline{\hat{A}_t} \equiv \langle \psi_H(t) | \hat{A}_H(t) | \psi_H(t) \rangle$

$$= \langle \psi(t) | \hat{U}(t,t_0)\, \hat{U}^{\dagger}(t,t_0)\, \hat{A}\, \hat{U}(t,t_0)\, \hat{U}^{\dagger}(t,t_0) | \psi(t) \rangle$$

$$= \langle \psi(t) | \hat{A} | \psi(t) \rangle$$

※ 两个绘景中的平均值重合！

- $i\hbar \frac{d}{dt} \overline{\hat{A}_t} = i\hbar \frac{d}{dt} \left[\langle \psi_H(t) | \hat{A}_H(t) | \psi_H(t) \rangle \right]$

$$= \langle \psi_H(t) | \left(i\hbar \frac{d}{dt} \hat{A}_H(t) \right) | \psi_H(t) \rangle$$

$$= \langle \psi_H(t) | [\hat{A}_H(t), \hat{H}_H(t)] | \psi_H(t) \rangle$$

若 $[\hat{A}_H(t), \hat{H}_H(t)] = 0$ ，

则称力学量 A 为守恒量 。

$$\text{좌변} \quad [\hat{A}_H(t),\ \hat{H}_H(t)]$$

$$= \hat{A}_H(t)\hat{H}_H(t) - \hat{H}_H(t)\hat{A}_H(t)$$

$$= \hat{U}^\dagger \hat{A} \underbrace{\hat{U}\hat{U}^\dagger}_{1} \hat{H}\hat{U} - \hat{U}^\dagger \hat{H} \underbrace{\hat{U}\hat{U}^\dagger}_{1} \hat{A}\hat{U}$$

$$= \hat{U}^\dagger \hat{A}\hat{H}\hat{U} - \hat{U}^\dagger \hat{H}\hat{A}\hat{U}$$

$$= \hat{U}^\dagger(t,t_0)\,[\hat{A},\hat{H}(t)]\,\hat{U}(t,t_0),$$

$$\text{우변} \quad [\hat{A}_H(t),\ \hat{H}_H(t)] = 0$$

$$\Rightarrow \quad [\hat{A},\hat{H}(t)] = 0.$$

$\hat{A}$ 가 보존량의 정의 증명 !

(d) A-表象

. $\hat{A}_H(t) = \hat{U}^\dagger(t\,t_0)\,\hat{A}\,\hat{U}(t,t_0)$

. $|a_i,t\rangle_H = \hat{U}^\dagger(t\,t_0)\,|a_i\rangle$

其中. $\boxed{\hat{A}\,|a_i\rangle = a_i\,|a_i\rangle}$

. $\hat{A}_H(t)\,|a_i,t\rangle_H = \hat{U}^\dagger(t\,t_0)\,\hat{A}\,\hat{U}(t\,t_0)\,\hat{U}^\dagger(t\,t_0)\,|a_i\rangle$

$= \hat{U}^\dagger(t\,t_0)\,\hat{A}\,|a_i\rangle = a_i\,\hat{U}^\dagger(t\,t_0)\,|a_i\rangle.$

$= a_i\,|a_i\,t\rangle_H$

即 $\boxed{\hat{A}_H(t)\,|a_i,t\rangle_H = a_i\,|a_i,t\rangle_H}$

. $\begin{cases} \int {}_H\langle a_i\,t\,|\,a_j\,t\rangle_H = \delta_{ij} & \text{(正交性)} \\ \sum_i |a_i\,t\rangle_H\,{}_H\langle a_i\,t| = \mathbb{1} & \text{(完备性)} \end{cases}$

基矢: $\{|a_i\,t\rangle_H\}$. 海森伯绘景下的

A-表象

- $|\psi_H(t)\rangle = \sum_i |a_i\, t\rangle_H \langle a_i\, t | \psi_H(t)\rangle$

* 海森伯绘景中，态不随时间变化，而基矢随时间变化！

例：谐振子相干态：求平均值 $\overline{\hat{x}_t}$.

· $\overline{\hat{x}_t} = \langle \psi_H(0) | \hat{x}_H(t) | \psi_H(0) \rangle$，设 $|\psi_H(0)\rangle = |\alpha\rangle$

· $\hat{H} = \frac{\hat{p}^2}{2m} + \frac{m\omega^2}{2}\hat{x}^2$ （不显含时间）

· $\hat{U}(t,t_0) = \exp\left[-\frac{i}{\hbar}\hat{H}(t-t_0)\right]$，

· 取 $t_0 = 0$，$\hat{U}(t) = \exp\left[-\frac{i}{\hbar}\hat{H}t\right]$，

解1：

由

$$\begin{cases} \hat{x}_H(t) = \exp\left[\frac{i}{\hbar}\hat{H}t\right]\hat{x}\exp\left[-\frac{i}{\hbar}\hat{H}t\right], \\ \hat{p}_H(t) = \exp\left[\frac{i}{\hbar}\hat{H}t\right]\hat{p}\exp\left[-\frac{i}{\hbar}\hat{H}t\right]. \end{cases}$$

海森伯方程：

$$\begin{cases} \frac{d}{dt}\hat{x}_H(t) = \frac{1}{i\hbar}\left[\hat{x}_H(t), \hat{H}_H(t)\right], & (\hat{x}_H(0) = \hat{x}) \\ \frac{d}{dt}\hat{p}_H(t) = \frac{1}{i\hbar}\left[\hat{p}_H(t), \hat{H}_H(t)\right]. & (\hat{p}_H(0) = \hat{p}) \end{cases}$$

初始条件

但是
$$[\hat{X}_H(t), \hat{H}_H(t)] = e^{\frac{i}{\hbar}\hat{H}t}[\hat{X}, \hat{H}]e^{-\frac{i}{\hbar}\hat{H}t}$$
$$= \frac{i\hbar}{m} e^{\frac{i}{\hbar}\hat{H}t}\hat{p}\, e^{-\frac{i}{\hbar}\hat{H}t}$$
$$= \frac{i\hbar}{m}\hat{P}_H(t) ,$$

类似，
$$[\hat{P}_H(t), \hat{H}_H(t)] = \overbrace{-m\omega^2}^{i\hbar}\hat{X}_H(t)$$

即
$$\begin{cases} \frac{d}{dt}\hat{X}_H(t) = \frac{\hat{P}_H(t)}{m} \\ \frac{d}{dt}\hat{P}_H(t) = -m\omega^2\hat{X}_H(t) \end{cases}$$

$$\Rightarrow \quad \frac{d^2}{dt^2}\hat{X}_H(t) = \frac{1}{m}\frac{d}{dt}\hat{P}_H(t)$$
$$= -\omega^2\hat{X}_H(t)$$

（经典力学方程：$\frac{d^2}{dt^2}x(t) = -\omega^2 x(t)$）

$$\begin{cases} \hat{X}_H(t) = \hat{C}_1 \cos\omega t + \hat{C}_2 \sin\omega t \\ \hat{P}_H(t) = m\frac{d}{dt}\hat{X}_H(t) = m\omega\left(\hat{C}_2\cos\omega t - \hat{C}_1 \sin\omega t\right) \end{cases}$$

初始条件：

$$\begin{cases} \hat{X}_H(0) = \hat{C}_1 = \hat{X} \\ \hat{P}_H(0) = m\omega\hat{C}_2 = \hat{P} \end{cases} \Rightarrow \begin{cases} C_1 = \hat{X} \\ C_2 = \dfrac{\hat{P}}{m\omega} \end{cases},$$

$$\Rightarrow \begin{cases} \hat{X}_H(t) = \hat{X}\cos\omega t + \dfrac{\hat{P}}{m\omega}\sin\omega t \\ \hat{P}_H(t) = \hat{P}\cos\omega t - m\omega\hat{X}\sin\omega t \end{cases}.$$

- 设 $t=0$ 时刻系统处于相干态 $|\alpha\rangle$ 中，求 t 时刻 $\hat{X}$ 的平均值。 $\rightarrow |\psi_H(0)\rangle = |\alpha\rangle$

$$\overline{\hat{X}_t} = \langle \psi_H(0) |\ \hat{X}_H(t)\ |\ \psi_H(t) \rangle = \langle \psi_H(0) |\ \hat{X}_H(t)\ |\ \psi_H(0) \rangle$$

$$= \langle \alpha |\ \hat{X}_H(t)\ |\ \alpha \rangle$$

$$= \langle \alpha | \left[\hat{x} \cos\omega t + \frac{\hat{p}}{m\omega} \sin\omega t \right] | \alpha \rangle$$

$$= \sqrt{\frac{\hbar}{2m\omega}} \cos\omega t \langle \alpha | (\hat{a}^\dagger + \hat{a}) | \alpha \rangle$$

$$+ i \sqrt{\frac{m\hbar\omega}{2}} \frac{1}{m\omega} \sin\omega t \langle \alpha | (\hat{a}^\dagger - \hat{a}) | \alpha \rangle$$

$$= \sqrt{\frac{\hbar}{2m\omega}} \cos\omega t \left(\alpha^* + \alpha \right)$$

$$+ i \sqrt{\frac{m\hbar\omega}{2}} \frac{1}{m\omega} \sin\omega t \left(\alpha^* - \alpha \right)$$

$$\left(\because \alpha = |\alpha| e^{i\phi} \right)$$

$$= \sqrt{\frac{\hbar}{2m\omega}} \cos\omega t\, |\alpha| \left(e^{-i\phi} + e^{i\phi} \right)$$

$$+ i \sqrt{\frac{m\hbar\omega}{2}} \frac{1}{m\omega} \sin\omega t\, |\alpha| \left(e^{-i\phi} - e^{i\phi} \right)$$

$$= 2|\alpha| \sqrt{\frac{E}{2m\omega}} \cos\omega t \cos\phi$$

$$+ 2|\alpha| \sqrt{\frac{E}{2m\omega}} \sin\omega t \sin\phi$$

$$= |\alpha| \sqrt{\frac{2E}{m\omega}} \cos(\omega t - \phi) .$$

即 $\hat{x}_t = |\alpha| \sqrt{\frac{2E}{m\omega}} \cos(\omega t - \phi)$,

与前述经典中得到的结果一致.

方法2：

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{m\omega^2}{2}\hat{x}^2 = \left(\hat{a}^+\hat{a} + \frac{1}{2}\right)\hbar\omega.$$

$$\overline{\hat{x}_t} = \langle\alpha|\ \hat{x}_H(t)\ |\alpha\rangle$$

$$= \langle\alpha|\ \sqrt{\frac{\hbar}{2m\omega}}\left(\hat{a}_H^+(t) + \hat{a}_H(t)\right)|\alpha\rangle.$$

运动方程：$\dfrac{d\hat{a}_H(t)}{dt} = \dfrac{1}{i\hbar}\left[\hat{a}_H(t),\ \hat{H}_H(t)\right]$.

但，$\dfrac{1}{i\hbar}\left[\hat{a}_H(t),\ \hat{H}_H(t)\right] = \dfrac{1}{i\hbar}\, e^{\frac{i}{\hbar}\hat{H}t}\left[\hat{a},\ \hat{H}\right] e^{-\frac{i}{\hbar}\hat{H}t}$

$$= \frac{1}{i\hbar}\, e^{\frac{i}{\hbar}\hat{H}t}\ \hbar\omega\hat{a}\ e^{-\frac{i}{\hbar}\hat{H}t}$$

$$= -i\omega\ \hat{a}_H(t), \quad \left(\hat{a}_H(t=0) = \hat{a}\right)$$

$$\Rightarrow \begin{cases} \hat{a}_H(t) = \hat{a}\, e^{-i\omega t} \\ \hat{a}_H^\dagger(t) = \hat{a}^+ e^{i\omega t} \end{cases}$$

$$\Rightarrow \overline{\hat{x}_t} = |\alpha|\sqrt{\frac{2\hbar}{m\omega}}\cos(\omega t - \phi).$$